

Kinematics in One and Two Dimensions, **solved**

Step-by-step solutions to every practice problem in Chapter 2. Try each problem on your own before you read its solution.

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P2.1

Yes. Velocity and acceleration are independent at any single instant. Acceleration measures how quickly the velocity is *changing*, not what it is.

Example: a ball thrown straight up. At the top of its flight $\mathbf{v} = \mathbf{0}$ for an instant, but the acceleration is still $\mathbf{g} = 9.8 \text{ m/s}^2$ downward. That is why the ball does not stay at the top. Another example is a car at the instant it starts from rest: $\mathbf{v} = \mathbf{0}$ while $\mathbf{a} > \mathbf{0}$.

P2.2

(a) **Yes.** Velocity is a vector. If the *direction* of motion changes while the magnitude stays fixed, the velocity changes but the speed does not. Example: a car going round a circular track at a steady 20 m/s. It is accelerating (towards the centre, as Chapter 4 shows) even though its speedometer reading is constant.

(b) **No.** Constant velocity means both magnitude and direction are constant, and the magnitude of the velocity *is* the speed. So the speed must also be constant.

P2.3

Set up. Take up as positive. The launch velocity is $+\mathbf{u}$, and the acceleration is $-\mathbf{g}$ throughout the flight, both going up and coming down.

- **v - t graph:** a straight line starting at $+u$, falling with slope $-g$, crossing $v = 0$ at $t = u/g$ (the top), and reaching $-u$ at $t = 2u/g$ (the catch).
- **a - t graph:** a horizontal line at $a = -g$ for the whole flight.

The slope of the v - t graph is $-g = -9.8 \text{ m/s}^2$, the same on the way up and the way down.

Check. The area under the v - t graph is a triangle above the axis followed by an equal triangle below it. They cancel, giving a total displacement of zero, as expected for a ball caught at its launch height. ✓

P2.4

Velocity is not zero. At the top, the vertical component is zero, but the horizontal component $u \cos \theta$ is unchanged. So $\vec{v} = u \cos \theta \hat{i}$, horizontal.

Acceleration is not zero. It is g downward everywhere along the path, including at the top.

At the highest point, velocity and acceleration are **perpendicular**. This is the only point on the path where that is true. The velocity is zero at the top only for a vertical launch ($\theta = 90^\circ$).

P2.5

The v - t graph is a trapezium. Find the area of each stage.

1. Accelerating: $s_1 = \frac{1}{2}(0 + 30)(10) = 150 \text{ m}$.
2. Constant speed: $s_2 = 30 \times 20 = 600 \text{ m}$.
3. Decelerating: $s_3 = \frac{1}{2}(30 + 0)(5.0) = 75 \text{ m}$.

Total distance: $s = 150 + 600 + 75 = 825 \text{ m}$ in a total time of $10 + 20 + 5 = 35 \text{ s}$.

$$\text{average speed} = \frac{825 \text{ m}}{35 \text{ s}} = 23.6 \text{ m/s}.$$

P2.6

Set up. Let the total fall time be T . The stone falls from rest, so the distance fallen after time t is $\frac{1}{2}gt^2$.

Execute. The distance fallen in the last second is the total distance minus the distance fallen in the first $T - 1$ seconds:

$$\frac{1}{2}gT^2 - \frac{1}{2}g(T-1)^2 = \frac{1}{2}g(2T-1) = 25.$$

$$2T - 1 = \frac{50}{9.8} = 5.102 \Rightarrow T = 3.05 \text{ s.}$$

Then the height of the tower is

$$H = \frac{1}{2}gT^2 = \frac{1}{2}(9.8)(3.051)^2 = 45.6 \text{ m.}$$

Check. In the first 2.05 s the stone falls $4.9(2.051)^2 = 20.6$ m, and $45.6 - 20.6 = 25.0$ m. ✓

P2.7

The acceleration depends on time, so the constant-acceleration equations do **not** apply. Integrate instead.

$$v(t) = v_0 + \int_0^t 6t \, dt = 2 + 3t^2 \text{ m/s.}$$

$$x(t) = \int_0^t (2 + 3t^2) \, dt = 2t + t^3 \text{ m.}$$

At $t = 2.0$ s:

$$v = 2 + 12 = 14 \text{ m/s,} \quad x = 4 + 8 = 12 \text{ m.}$$

P2.8

Set up. Put the origin at the police car and start the clock as the speeder passes.

- Speeder: $x_s = 30t$.
- Police car: $x_p = \frac{1}{2}(3.0)t^2 = 1.5t^2$.

Execute. The police car catches up when $x_p = x_s$:

$$1.5t^2 = 30t \Rightarrow t(1.5t - 30) = 0 \Rightarrow t = 20 \text{ s.}$$

The root $t = 0$ is the moment the speeder passes.

Distance travelled: $x = 30(20) = 600$ m. Speed of the police car: $v = at = 3.0(20) = 60$ m/s.

Evaluate. The police car is going exactly twice the speeder's speed when it catches up. That is always true here: both cars cover the same distance in the same time, so they have the same average velocity, and the police car's average velocity is half its final velocity.

P2.9

Set up. Put the origin at the launch point, with y up. Then $u_x = 20 \cos 30^\circ = 17.32$ m/s and $u_y = 20 \sin 30^\circ = 10.0$ m/s. The ground is at $y = -25$ m.

Vertical motion:

$$-25 = 10t - 4.9t^2 \Rightarrow 4.9t^2 - 10t - 25 = 0.$$

$$t = \frac{10 + \sqrt{100 + 4(4.9)(25)}}{9.8} = \frac{10 + \sqrt{590}}{9.8} = \frac{10 + 24.29}{9.8} = 3.50 \text{ s.}$$

Horizontal motion:

$$x = u_x t = 17.32 \times 3.50 = 60.6 \text{ m.}$$

Evaluate. Do *not* use $R = u^2 \sin 2\theta / g$ here. That would give 35.3 m, because it assumes the ball lands at its launch height.

P2.10

Angles. From $R = u^2 \sin 2\theta / g$:

$$\sin 2\theta = \frac{Rg}{u^2} = \frac{60(9.8)}{900} = 0.6533.$$

So $2\theta = 40.8^\circ$ or $180^\circ - 40.8^\circ = 139.2^\circ$, giving

$$\theta_1 = 20.4^\circ, \quad \theta_2 = 69.6^\circ.$$

These are complementary angles, as expected.

Heights. From $H = u^2 \sin^2 \theta / 2g$:

$$H_1 = \frac{900 \sin^2 20.4^\circ}{19.6} = \frac{900(0.1215)}{19.6} = 5.6 \text{ m}, \quad H_2 = \frac{900 \sin^2 69.6^\circ}{19.6} = \frac{900(0.8785)}{19.6} = 40.3 \text{ m}$$

Check. For complementary angles,

$$H_1 H_2 = \frac{u^4 \sin^2 \theta \cos^2 \theta}{4g^2} = \frac{R^2}{16} = \frac{3600}{16} = 225 \text{ m}^2,$$

and $5.58 \times 40.34 = 225. \checkmark$

P2.11

Set up. Let the man walk in the $+x$ direction at 4.0 km/h, and let the rain fall at speed v_r in the $-y$ direction.

Execute. The velocity of the rain relative to the man is

$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m = -4.0 \hat{i} - v_r \hat{j}.$$

It makes 30° with the vertical, so

$$\tan 30^\circ = \frac{4.0}{v_r} \Rightarrow v_r = \frac{4.0}{0.577} = 6.9 \text{ km/h}.$$

The speed relative to the man is

$$|\vec{v}_{rm}| = \frac{4.0}{\sin 30^\circ} = 8.0 \text{ km/h}.$$

P2.12

Convert the speed: $u = 54 \text{ km/h} = 15 \text{ m/s}$. For each stop use $a = u^2/2s$, from $v^2 = u^2 + 2as$ with $v = 0$.

(a) Passenger compartment, stopping in $s = 0.60 \text{ m}$:

$$a = \frac{15^2}{2(0.60)} = 188 \text{ m/s}^2 = \frac{188}{9.8} g = 19 g.$$

Duration, from $s = \frac{1}{2}(u + 0)t$:

$$t = \frac{2s}{u} = \frac{2(0.60)}{15} = 0.080 \text{ s}.$$

(b) Unbelted passenger, stopping in $s = 0.050 \text{ m}$:

$$a = \frac{225}{2(0.050)} = 2250 \text{ m/s}^2 \approx 230 g.$$

(c) Belted passenger with an airbag, stopping in $s = 0.90$ m:

$$a = \frac{225}{2(0.90)} = 125 \text{ m/s}^2 \approx 13g.$$

Evaluate. The seat belt and airbag cut the deceleration by a factor of about 18 compared with hitting the dashboard: from a likely fatal level to a survivable one. Notice that the car's crumple zone counts in the restrained passenger's stopping distance. The passenger rides the car down. This is why modern cars are designed to deform, rather than to be as rigid as possible.

P2.13

Set up. Use axes along the incline (x' , up the slope) and perpendicular to it (y'). The launch velocity makes angle $(\theta - \alpha)$ with the incline, and gravity has components along both axes:

- Along the incline: $u_{x'} = u \cos(\theta - \alpha)$ and $a_{x'} = -g \sin \alpha$.
- Perpendicular to the incline: $u_{y'} = u \sin(\theta - \alpha)$ and $a_{y'} = -g \cos \alpha$.

Time of flight. The projectile lands when $y' = 0$:

$$u \sin(\theta - \alpha) T - \frac{1}{2} g \cos \alpha T^2 = 0 \Rightarrow T = \frac{2u \sin(\theta - \alpha)}{g \cos \alpha}.$$

Range along the incline.

$$R = u \cos(\theta - \alpha) T - \frac{1}{2} g \sin \alpha T^2.$$

Substitute T and factor out $\frac{2u^2 \sin(\theta - \alpha)}{g \cos^2 \alpha}$:

$$R = \frac{2u^2 \sin(\theta - \alpha)}{g \cos^2 \alpha} \left[\cos(\theta - \alpha) \cos \alpha - \sin(\theta - \alpha) \sin \alpha \right].$$

The bracket is $\cos((\theta - \alpha) + \alpha) = \cos \theta$, so

$$R = \frac{2u^2 \cos \theta \sin(\theta - \alpha)}{g \cos^2 \alpha}. \checkmark$$

Maximising. Use the product-to-sum identity $2 \sin(\theta - \alpha) \cos \theta = \sin(2\theta - \alpha) - \sin \alpha$:

$$R = \frac{u^2 [\sin(2\theta - \alpha) - \sin \alpha]}{g \cos^2 \alpha}.$$

This is largest when $\sin(2\theta - \alpha) = 1$, that is, $2\theta - \alpha = 90^\circ$:

$$\theta_{\text{best}} = 45^\circ + \frac{\alpha}{2}.$$

The best direction bisects the angle between the incline and the vertical. The maximum range is

$$R_{\text{max}} = \frac{u^2(1 - \sin \alpha)}{g \cos^2 \alpha} = \frac{u^2(1 - \sin \alpha)}{g(1 - \sin \alpha)(1 + \sin \alpha)} = \frac{u^2}{g(1 + \sin \alpha)}.$$

Check. With $\alpha = 0$ these results reduce to $\theta = 45^\circ$ and $R_{\text{max}} = u^2/g$, the level-ground results. ✓

P2.14

(a) The acceleration depends on velocity, and we want v as a function of x , so use $a = v dv/dx$:

$$v \frac{dv}{dx} = -kv^2 \Rightarrow \int_{v_0}^v \frac{dv}{v} = -k \int_0^x dx \Rightarrow v = v_0 e^{-kx}.$$

(b) For $v(t)$, use $dv/dt = -kv^2$:

$$\int_{v_0}^v \frac{dv}{v^2} = -k \int_0^t dt \Rightarrow -\frac{1}{v} + \frac{1}{v_0} = -kt \Rightarrow v = \frac{v_0}{1 + kv_0 t}.$$

(c) The speed halves when $e^{-kx} = \frac{1}{2}$:

$$x_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693}{k}.$$

Evaluate. The speed halves over every further distance $\ln 2/k$, like a half-life measured in metres. So the body never stops in a finite distance. Compare Example 2.5, where $a = -kv$ gave a finite stopping distance.

P2.15

Set up. With up positive and launch from the ground:

$$h = ut - \frac{1}{2}gt^2 \Rightarrow \frac{1}{2}gt^2 - ut + h = 0.$$

The times t_1 and t_2 at which the ball is at height h are the two roots of this quadratic.

Execute. For a quadratic $At^2 + Bt + C = 0$, the product of the roots is C/A and their sum is $-B/A$:

$$t_1 t_2 = \frac{h}{g/2} = \frac{2h}{g} \Rightarrow h = \frac{1}{2} g t_1 t_2. \checkmark$$

$$t_1 + t_2 = \frac{u}{g/2} = \frac{2u}{g} \Rightarrow u = \frac{1}{2} g (t_1 + t_2). \checkmark$$

Evaluate. By symmetry the top of the flight is at the midpoint time $\frac{1}{2}(t_1 + t_2) = u/g$, which is consistent with $t_{\text{up}} = u/g$. $\checkmark$

P2.16

Identify. The current (5.0 m/s) is faster than the boat (3.0 m/s), so *some* drift is unavoidable. We need the heading that minimises it.

Set up. Let the boat head at angle α upstream from the straight-across direction. Relative to the ground:

- Speed across: $v \cos \alpha$, with $v = 3.0$ m/s.
- Speed downstream: $u - v \sin \alpha$, with $u = 5.0$ m/s.

The crossing time is $t = d/(v \cos \alpha)$, so the drift is

$$x(\alpha) = \frac{d(u - v \sin \alpha)}{v \cos \alpha}.$$

Execute. Set $dx/d\alpha = 0$. By the quotient rule, the numerator of the derivative must vanish:

$$-v \cos^2 \alpha + (u - v \sin \alpha) \sin \alpha = 0 \Rightarrow u \sin \alpha = v(\cos^2 \alpha + \sin^2 \alpha) = v.$$

$$\sin \alpha = \frac{v}{u} = \frac{3.0}{5.0} = 0.60 \Rightarrow \alpha = 36.9^\circ \text{ upstream of straight across.}$$

That is 53.1° from the upstream bank.

With $\cos \alpha = 0.80$:

- Crossing time: $t = \frac{120}{3.0 \times 0.80} = 50$ s.
- Minimum drift: $x = (5.0 - 3.0 \times 0.60) \times 50 = 3.2 \times 50 = 160$ m.

In general, $x_{\min} = d\sqrt{u^2 - v^2}/v = 120 \times 4/3 = 160$ m. $\checkmark$

Evaluate. Heading straight across would take 40 s and give 200 m of drift. Aiming upstream costs 10 s but saves 40 m. Geometrically, the boat's ground velocity is then perpendicular to its water velocity.

P2.17

Stage 1: powered flight. With $a = 20 \text{ m/s}^2$ for $t = 10 \text{ s}$, starting from rest:

$$v_1 = at = 200 \text{ m/s}, \quad h_1 = \frac{1}{2}at^2 = \frac{1}{2}(20)(100) = 1000 \text{ m}.$$

Stage 2: coasting upward. Now $a = -g$. The rocket rises a further

$$\Delta h = \frac{v_1^2}{2g} = \frac{200^2}{19.6} = 2041 \text{ m},$$

taking $t_2 = v_1/g = 200/9.8 = 20.4 \text{ s}$.

(a) At burnout: speed 200 m/s, height 1000 m.

(b) Maximum height: $H = 1000 + 2041 = 3041 \text{ m} \approx 3.04 \text{ km}$.

Stage 3: free fall from rest at height H .

$$t_3 = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2(3041)}{9.8}} = 24.9 \text{ s}.$$

(c) Total time: $10 + 20.4 + 24.9 = 55.3 \text{ s}$.

Evaluate. The rocket keeps rising for twice as long after burnout as it did under power. Its speed at impact is $\sqrt{2gH} = 244 \text{ m/s}$, faster than at burnout, because it falls from three times the burnout height. Real rockets are slowed considerably by air resistance and carry parachutes.

P2.18

Convert the speed: $v = 50 \text{ km/h} = 13.9 \text{ m/s}$.

(a) **Minimum stopping distance** = reaction distance + braking distance:

$$d_{\text{stop}} = vt_r + \frac{v^2}{2a} = 13.9(1.0) + \frac{(13.9)^2}{2(4.0)} = 13.9 + 24.1 = 38.0 \text{ m}.$$

A driver closer than 38.0 m to the stop line when the light turns yellow cannot stop in time.

(b) **Clearing.** The worst case is a driver just under 38.0 m from the line. To clear, the rear of the car must pass the far side of the intersection, so the car must cover

$$d_{\text{clear}} = 38.0 + 20 + 5.0 = 63.0 \text{ m}$$

at 13.9 m/s, which takes

$$t = \frac{63.0}{13.9} = 4.5 \text{ s.}$$

So the yellow phase (together with any all-red interval) must last at **least 4.5 s**. If it were shorter, there would be a band of distances in which a driver could neither stop nor clear: the dilemma zone.

Evaluate. Traffic engineers use exactly this kinematic model, usually splitting the time into a yellow interval of about 3–5 s plus an “all-red” clearance interval. A higher speed limit needs a longer yellow, because the braking distance grows as v^2 .

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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